

Assignment

1) Explain the programming and python in detail.

Defination:-

Programming is the process of writing instructions that a computer follows to performs a task.

Purpose:-

- i) They only understand instructions
- ii) programmes help automatic tasks & solve problems.
- iii) computers cannot think on their own.

→ **Comments** used to explain code & non executable lines used for explanation.

→ Types in comments

1) Single line Comment

Syntax:- # This is single line comment

2) multi-line comment

Syntax:- " This is
multiline
comment"

Importance of Python:-

- platform Independent
- large library support
- simple & easy to learn
- High level interpreted language.

2)

2) Describe Data types & operators in python with suitable Examples.

Data types specifies what kind of data variable holds & Determines data type automatically.

Datatypes are classified into

1) Complex / Numeric data type:

→ int

* store whole numbers

* Positive or negative

a = 10

b = -8

→ float

* stores decimal numbers

m = 7.14

n = 0.9

→ Complex

* stores numbers in (a+bi) format

z = 2+3j

2) Text Data type:-

→ stores text (or) characters

→ written inside quotes

Name = "Zebra"

num = "one"

3) Boolean Data types

→ bool

→ stores only True or False.

→ used in decision making

is_vaild = True

4) Sequence Data Types

→ List

• ordered collection

• mutable (change)

grades = [75, 96, 84]

→ Tuple

• ordered collection

• immutable (cannot change)

Shades = (maroon, beige, gold)

5) Set data type

→ set

• unordered collection

• No duplicate values

S = {1, 2, 3}

Mapping data types

→ dict

- stores data in key value pairs

```
student = {"name":
```

```
"Srujana", "age": 20 }
```

→ Identification checked using type () function

```
x = 9  
Print (type(x))
```

→ Various Python operators are categorized in following.

1) Arithmetic operator:-

- * These are used to perform mathematical calculations & numbers like Integers

Operators

+ → addition

- → subtraction

* → Multiplication

/ → Division

% → modulus (remainder)

// → floor division

** → Exponent (power)

(Q) Comparison Operators:-

- These Operators used to compare 2 values & result always a boolean value (T or F)

Operators

= → Equal to

! → Not equal

> → Greater than

< $\Rightarrow$ less than

>= $\Rightarrow$ greater than or equal

<= $\Rightarrow$ less than or equal

$\rightarrow$ usage of operators in real world.

- Total bill, tax

- Average score

- Age verification

- Filtering data.

3) Explain python input & output operations in detail.

$\rightarrow$ Input is the data given to a program by the user. Input & output help in making programs interactive & dynamic.

- Input () function

Basic syntax $\rightarrow$ variable = input ("message")

Ex:- name = input ("name")

print ("Hello, name")

$\rightarrow$ Type of conversion while taking input

- Since input is always string, so conversion is required.

$\rightarrow$ Integer input

age = int (input ("Enter age:"))

print age.

$\rightarrow$ float input.

Price = float (input ("Enter price:"))

print (price)

$\rightarrow$ Taking multiple inputs

using separate statements

- 2)
 $<$ $\Rightarrow$ less than
 $>=$ $\Rightarrow$ greater than or equal
 $<=$ $\Rightarrow$ less than or equal

$\rightarrow$ usage of operators in real world.

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- Average score
- Age verification
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- Input () function

Basic Syntax $\rightarrow$ Variable = input ("message")

Ex:- name = input ("name")
Print = ("Hello, name")

$\rightarrow$ Type of conversion while taking input

- Since input is always string, so conversion is required.

$\rightarrow$ Integer input

age = int (input ("Enter age;"))

Print age.

$\rightarrow$ Float input.

Price = float (input ("Enter price;"))

Print (price)

$\rightarrow$ Taking multiple inputs
using separate statements

(2)

```
a=int(input("Enter a:"))  
b=int(input("Enter b:"))
```

Using split()

```
a,b=map(int,input("Enter two numbers:"))  
Print(a,b)
```

→ Formatted output can be followed as :-

- Using f-strings

```
name="Riya"
```

```
age=24
```

```
Print(f"name : {name} Age : {age}")
```

- input & output with expressions

```
a=int(input("Enter number:"))
```

```
Print("Square :", a*a)
```

- Custom separator & end

```
Print("html" "css" sep=" ",)
```

```
Print("Hello", end=" ")
```

```
Print("mine")
```

4) Discuss control statement & decision-making statement in Python.

→ Control statements

- Control statements are used to control the flow of execution of a Python program.

Types of control statements

→ Decision Making statements

→ Looping statements

→ Jump / control transfer.

→ Decision Making statements are used to execute code based on conditions.

i) if statement

• Execute block of code only if the condition is true if it is false

Syntax

if condition:
statement

Ex:- age = 17

if age >= 18

Print ("Eligible to vote")

ii) if-else statement

• Execute one block if condition is true, executes another block if condition is false.

Syntax:

if condition
statement

Ex:-

num = 5

if num % 2 == 0

Print ("Even number")

else

Print ("Odd number")

iii) if-else statements-

• used to check multiple conditions

• elif means "else if"

Syntax:

if condition:
statement

elif condition (2)
statement

elif condition (3)

else;
statement

ex:

num = 9

if num > 0;

Print ("positive number")

elif num < 0;

Print ("negative number")

else

Print ("zero")

3) Logical operators

• Logical operators used to combine multiple conditions, used in decision making statements

Operator

- and → True if both are true
- or → True if anyone is true
- not → reverse the result.

4) Assignment Operators

• These operators are used to assign (or) update.

Operator	Ex
=	2 = 10
+=	2 += 3
-=	2 -= 2
*=	2 * = 2
/=	2 / = 2

5) Membership Operators

• This operators check wheater a value : exist in a sequence used with lists, string, tuples

Operator	
in	Present
not in	not present.

6) Identify Operators

• Identify operators check whether 2 variables refers to same memory object, not compare values

Operator	
is	Same object
is not	Different object.

5) Write an essay on python programming fundamentals.

- Programming helps in automating repetitive tasks improving accuracy & saving time. it involves breaking down a complex problem into smaller steps & performs each step logically.
- The simplicity allows programmers to focus more on problem solving than syntax rules few lines compared to other language
- Data types, operators & input/output operations to store data. Perform operations like arithmetic, relational & logical operations.
- Decision making statement as if, else if & control flow statement order program statement are executed. Decision making help in implementing different actions.

Python helpful language for beginner & professional - Python solve problems logically.

Real-World problems Using Python

1) Movie ticket pricing.

A movie theater charges

₹150 - children (age ≤ 13)

₹250 - adults (age 13-59)

₹200 - seniors (age ≥ 60)

→

```
# Movie ticket pricing
```

```
age = int(input("Age"))
```

```
is3D = int(input("3D"))
```

```
if is3D == 1:
```

```
    price = 50
```

```
    if age  $\leq 13$ :
```

```
        price = 150
```

```
    elif age  $\leq 60$ :
```

```
        price = 250
```

```
    print(price).
```

2) college attendance rule

```
attper = int(input("Enter your attendance Percentage"))
```

```
medlex = int(input("Enter 1 if you have medical  
Certificate else 0:"))
```

```
if attper  $\geq 75$  or (attper  $\geq 60$  and Medlex == 1):
```

```
    print("you allowed to write Exam")
```

```
else:
```

```
    print("you not allowed to write Exam")
```

3) Program On E-commerce Discount.

```
bilamo = int(input("Enter your bill amount:"))
```

```
isprime = int(input("Enter 1 if you are prime  
member else 0:"))
```

```

if is prime == 1:
    if bilamo >= 5000:
        dis = 0.25 * bilamo
    elif 2000 <= bilamo <= 4999:
        dis = 0.15 * bilamo
    else:
        dis = 0
else:
    if bilamo >= 5000:
        dis = 0.20 * bilamo
    elif 2000 <= bilamo <= 4999:
        dis = 0.10 * bilamo
    else:
        dis = 0
amo = bilamo - dis
Print ("final amount to be paid is", amo).

```

4) Program On Smartphone Battery warning.

```

batper = int (input ("Enter your batter percentage"))
ischar = int (input ("Enter 1 if phone is charging
                    else 0"))
if ischar == 0:
    if batper <= 20:
        Print ("Low battery")
    elif 21 <= batper <= 80:
        Print ("Normal")
    else:
        print ("full")
elif ischar == 1:
    Print ("charging")
else:
    print ("Enter 0 or 1")

```

Program on Driving License check

```
age = int (input ("Enter your Age:"))  
test passed = int (input ("Enter 1 if you passed the test else 0:"))  
if (age >= 18 and test passed == 1) Or age >= 60:  
    Print ("Eligible")  
else:  
    Print ("Not Eligible")
```

6) Program on online food delivery.

```
amount = int (input ("Enter Order amount"))  
is hold = int (input ("Enter 1 if you are a gold member else 0:"))  
dis = int (input ("Enter distance in km:"))  
if (amount >= 500 and dis < 10) Or (is hold == 1 and dis < 10):  
    Print ("Free delivery")  
else:  
    Print ("Delivery is never free")
```

7) Program on Bank loan Approval.

```
Sal = int (input ("Enter your salary:"))  
CERSO = int (input ("Enter your Credit score"))  
if (Sal >= 3000 & CERSO >= 700) Or Sal > 50000:  
    Print ("Loan Approved")  
else:  
    Print ("Loan Rejected")
```


8) Program on Electricity Bill

```
units = int(input("Enter number of units consumed"))
```

```
if units <= 100:
```

```
    bill = units * 2
```

```
elif units <= 200:
```

```
    bill = 100 * 2 + (units - 100) * 3
```

```
else:
```

```
    bill = 100 * 2 + 100 * 3 + (units - 200) * 5
```

```
Print("Final bill amount: ₹, bill):
```

9) Program on Student Scholarship

```
marks = int(input("Enter your marks"))
```

```
faminc = int(input("Enter family income"))
```

```
issinpar = int(input("Enter 1 if you have single parent else 0:"))
```

```
if issinpar == 1 & marks >= 85:
```

```
    Print("You are eligible for a scholarship")
```

```
elif marks >= 85 & faminc < 50000:
```

```
    Print("You are eligible for a scholarship")
```

```
else:
```

```
    Print("You are not eligible for a scholarship")
```

10) Program on online Exam Result.

```
theory = int(input("Enter your theory marks"))
```

```
Practical = int(input("Enter your practical marks:"))
```

```
total = theory + Practical
```

```
if total >= 100 or (theory >= 40 & Practical >= 40):
```

```
    Print("Pass")
```

```
else:
```

```
    Print("Fail")
```

Hotel room pricing

A Hotel charges

₹ 3000 Per day for normal days

₹ 4000 per day on weekends

print final bill

is weekend = int(input("week"))

days = int(input("Days"))

if weekend == 1

cost = 4000

else:

cost = 3000

total = cost * days

if days > 3:

total = total * 0.15

Print (total)

12) Gaming level unlock

A Game unlocks next level if:

Score ≥ 100 of Player has a premium Pass

But if Player used cheating access is denied

Input: score, is premium, used cheat

Score = int(input("score"))

Premium = int(input("premium"))

cheat = int(input("cheat"))

if cheat == 1:

Print ("Access Denied")

elif score ≥ 100 or Premium == 1

Print ("level unlocked")

else

Print ("level locked").

13) Mobile data usage:

A network gives unlimited data if:
daily wage ≤ 200 (or) user has unlimited plan
But if roaming is ON, Unlimited Plan forming

```
data = float(input("Data"))  
unlimited = int(input("Unlimit"))  
roaming = int(input("Roam"))  
if roaming == 1:  
    print("limited Data")  
elif data <= 200 or unlimited == 1:  
    print("unlimited Data")  
else:  
    print("limited Data")
```

14) Office entry system.

An employee can enter the office if:

ID card is valid AND

(fingerprint matches OR faces can match)

But if it is a holiday, entry is denied for everyone

```
id = int(input("ID"))  
finger = int(input("finger"))  
face = int(input("face"))  
holiday = int(input("holiday"))  
if holiday == 1:  
    print("entry denied")  
elif id == 1 and (finger == 1 or face == 1):  
    print("Entry Allowed")  
else:  
    print("Entry Denied")
```

Movie Rating Display.

A movie app shows rating based on average score:

Avg $\geq 8.5 \rightarrow$ "Excellent"

Avg b/w 6.0 & 8.4 "Good"

Avg $\leq 6.0 \rightarrow$ "Average"

But if the movie is marked as editor "choice" always show "Recommended"

Print the message

```
 $\rightarrow$  rating = float (input("Rating"))
```

```
editor = int (input ("editor"))
```

```
if editor == 1:
```

```
    print ("Recommended")
```

```
elif rating  $\geq$  8.5
```

```
    print ("Excellent")
```

```
elif rating  $\geq$  6
```

```
    print ("Good")
```

```
else:
```

```
    print ("Average")
```